MILITARY SPECIFICATION
DYNAMITE (COMMERCIAL)

1. SCOPE

1.1 This specification covers the basic requirements for four types of commercial dynamite that shall be of a non-freezing type and shall contain no non-combustible material other than a commercially acceptable antacid.

1.2 Classification. The dynamite shall be of the following types (see 3.2 and 6.1):

Type I – (see 3.2 Table I)
Type II – (see 3.2 Table I)
Type III – (see 3.2 Table I)
Type IV – (see 3.2 Table I)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-13638 – Dynamite (Ammonia – Gelatin, and Gelatin types), Packaging of.

STANDARDS

MILITARY

MIL-STD-105 – Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 – Quality Assurance Terms and Definitions.
MIL-STD-1168 – Lot Numbering of Ammunition.
MIL-STD-1235 – Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Unless otherwise specified types I, II and IV dynamite shall be furnished as sticks 1¼ inches in diameter by 8 inches long and type II dynamite shall be furnished either as sticks 1¼ inches in diameter by 8 inches long or as sticks 2¾ inches in diameter by 16 inches long. The paper shells for all types of dynamite shall be sprayed with paraffin before loading. The loaded paper shells for type I and II dynamite shall be either dipped in molten paraffin or sprayed with paraffin. As an alternative to dipping or spraying, the loaded paper shells for type I and II dynamite may be packed using a polyethylene case liner (see 4.3.1 for test procedure).

3.2 Composition. The composition of the dynamite shall be as specified in Table I, for the applicable type dynamite. The manufacturer shall certify that the percentages of the ingredients contained in the composition complies with that stated in Table I.

3.3 Moisture content. The moisture content of the dynamite shall not exceed 2 percent, when determined as specified in 4.3.2.

3.4 Stability. The dynamite shall cause no discoloration of the potassium iodide starch test paper for a minimum period of 10 minutes, when tested as specified in 4.3.3.

FSC: 1375

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TABLE I

Ingredient	Percent Type			
	I	II	III	IV
Explosive oil	16 + 4[a]	25 + 5	50 + 4	64 + 4
Ammonium nitrate	62 + 4	13 + 4	–	–
Sodium nitrate	10 + 4	47 + 5	39 + 5	28 + 4
Carbonaceous combustible matter and antacid	(Manufacturers option) [b]			

[a] May be nitrostarch

[b] The choice as to type or amount of carbonaceous combustible material and antacid used in the manufacture of the dynamite, shall be left to the discretion of the manufacturer.

3.5 Rate of detonation. The rate of detonation shall be as specified in Table II for the applicable type dynamite, when determined as specified in 4.3.4.

3.6 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with the best standard practice.

TABLE II

Test	Type			
	I	II	III	IV
Rate of detonation, thousands of feet per second, minimum	10	12	16	18

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

4.1.1 *Contractor quality assurance system.* If the contractor desires to utilize a quality assurance system which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment), and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure which affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 *Submission of product.* At the time the complete lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate

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which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all requirements and provisions of the approved current written description of the material.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all material used directly by the contractor when such material is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure (see 4.1.1), this detail specification and other contractual documents as a guide, the Government shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with (a) or (b) as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality provisions of the other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, resampling, re-instruction of the supplier's employees, or other appropriate action:

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

b. Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 *Surveillance.* Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determinations that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 *Product inspection.* Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions

4.2.1 *Lot formation.* A lot shall consist of one particular type of dynamite produced by one manufacturer in one unchanged process, in accordance with the same specification and same specification revision. Process changes not affecting performance, safety, interchangeability, or storage, as determined by the Government, shall not necessitate changing the lot interfix number. The product shall be submitted for inspection in accordance with Standard MIL-STD-105, MIL-STD-1235, as applicable. The criteria and procedure for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.2.2 *Sampling*

4.2.2.1 For test specified in 4.3.1, the sample shall be obtained by removing 50 sticks of dynamite from 50 different cases. The sticks of dynamite shall be obtained by selecting one stick of dynamite from

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50 individual shipping containers selected randomly from the day's production. If this sample fails to comply with any of the requirements specified in 3.1, the day's production shall be rejected.

4.2.2.2 For tests specified in 4.3.2 to 4.3.4, the sample shall be obtained from a batch produced that day by removing 12 sticks from that batch. Randomly select two sticks from the 12 and determine the moisture content (4.3.2) and stability (4.3.3) of the composition contained therein by removing explosive composition from the two sticks and mixing the composition before performing tests. If this sample fails to comply with any of the requirements specified in 3.3 or 3.4, the day's production shall be rejected.

4.2.2.3 For test specified in 4.3.4, select three sticks from those remaining after performing tests specified in 4.2.2.2 and perform the rate of detonation test. If this sample fails to comply with any of the requirements specified in 3.5, the day's production shall be rejected. (NOTE: The tests depicted in 4.2.2.1, 4.2.2.2, and 4.2.2.3 shall be performed each day that the product is being produced.)

4.3 Test methods and procedures

4.3.1 *Description, Code No. 01001.* Visually examine the 100 sticks of dynamite obtained per 4.2.2.1 to determine that they meet the requirements specified in 3.1 and/or the contract.

4.3.2 *Moisture content, Code No. 02001.* Determine the moisture content of the sample in accordance with a suitable method that appears in a standard or textbook reference and which was previously approved by the contracting officer.

4.3.3 *Stability test, Code No. 03001.* The stability test shall be conducted as follows:

Transfer 2-gram portions of the explosive in its original condition, without preliminary drying or preparation other than thorough mixing, to each of three test tubes which are 5.5 inches long, 0.5 inch internal diameter, and 0.62 inch external diameter, taking care that no material comes in contact with the side walls of the test tubes. Each tube shall be provided with a closure consisting of a clean, tightly fitting cork stopper, through which passes a tightly fitted glass rod equipped with a platinum wire holder for the test paper strip. Standard potassium iodide starch paper, about one inch long and 3/8 inch wide, shall be used. The upper portion of the paper strip shall be moistened with a 50-percent solution of pure glycerin in water, using a small glass rod with which to apply the solution. The test paper assembly shall be inserted into the test tube so that the lower edge of the test paper is suspended approximately 2.5 inches above the test material. As a control on the cleanliness of the equipment and as a basis for the differentiation of color at the end point, a blank assembly shall be prepared, similar to the test assembly but without the explosive. The heat test bath shall be regulated at 82.2 degrees (°) Centigrade (C.) ± 1.0°C., the tubes inserted to a depth of 2.5 inches, and the time noted. The bath shall be placed so that the tubes are viewed against a white background illuminated by bright, diffused daylight. If a slight film of moisture condenses on the inside of the tubes as the test progresses, the lower line of demarcation between the wet and dry portions of the test paper shall be kept in line with the moisture film's lower edge. Observations shall be made at one minute intervals; continuous observation should be avoided. The first appearance of discoloration at the lower line of demarcation on the test paper in any tube marks the end of the test. Confirmation of the presence of this discoloration is made by comparison with the blank assembly. This minimum time required for the completion of the test shall represent the stability test value of the sample dynamite, to determine compliance with the specified requirement. For routine testing, all glassware shall be thoroughly washed in warm water, using soap and brush, rinsed in distilled water and dried. New corks shall be used for each test. The standard potassium iodide starch test papers shall be furnished by the contracting agency. (NOTE: An alternative method found applicable and acceptable to the contracting officer may be used.)

4.3.4 *Rate of detonation, Code No. 04001.* The rate of detonation of the dynamite shall be determined by the D'Autriche Method or any alternative method found applicable and acceptable to the contracting officer, to ascertain compliance with the specified requirement.

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5. PREPARATION FOR DELIVERY

5.1 The material described in this specification shall be prepared for delivery in accordance with specification MIL-D-13638.

6. NOTES

6.1 Intended use. Commercial dynamite is intended for the following uses:

Type I – For use in general blasting of rocks and stumps, and ditching in dry areas by individual initiation of each stick.

Type II – For use in blasting of very hard rocks and general blasting of rocks, stumps and earth in wet areas.

Type III – For use in submarine blasting, cutting of trees, shearing of light steel and ditching in wet areas by self-propagation.

Type IV – For use in heavy duty submarine blasting, cutting of large trees and shearing heavy steel.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification. In addition, the procurement document shall specify the type of dynamite required and packaging required.

6.3 Inspection code numbers. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Custodian:

Army—MU

PROJECT Number: 1375-A973

Preparing activity:

Army—MU

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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